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Abstract 13967: Development of Pharmacogenomic Algorithm for Warfarin Dose Monitoring in Indian Patients

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Abstract

Introduction: Wide inter-individual variation in the dose of warfarin has become a serious cause of concern, which is attributable to physiological [age, body mass index, gender] and genetic factors [SNPs in cytochrome P450 subfamily IIC polypeptide9 (CYP2C9) and vitamin K epoxide reductase complex 1 (VKORC1) genes]. In view of ethnic/population based differences in the distribution of these SNPs, we planned to develop ethnic group/population-specific pharmacogenomic algorithm, which is first of its kind in India in the arena of "personalized medicine".

Methods: Subjects (n=120) receiving warfarin were screened for 5 genetic polymorphisms i.e. CYP2C9*2, CYP2C9*3, VKORC1*3, VKORC1*4, VKORC1 D36Y using PCR-RFLP methods. A pharmacogenomic algorithm was developed using multiple linear regression analysis. Actual dose of warfarin was correlated with predicted dose in three groups: subjects who attained therapeutic target of international normalized ratio (INR) [Derivation cohort], subjects with INR below/above the therapeutic target of INR [Validation cohort].

Results: The allele frequencies of CYP2C9*2, CYP2C9*3, VKORC1*3, VKORC1*4, VKORC1 D36Y in the study population were 7.95%, 8.43%, 12.78%, 7.22%, 0% respectively. Using the data from Derivation cohort, we have deduced the following pharmacogenomic algorithm: Warfarin dose (mg/week) = $-0.07988 (\text{age}) + 6.88535 (\text{gender}) - 0.09913 (\text{BMI}) + 6.59161 (\text{CYP2C9*2}) - 6.83775 (\text{CYP2C9*3}) - 1.89090 (\text{VKORC1*3}) + 3.68511 (\text{VKORC1*4}) + 34.26665$. Actual dose of warfarin correlated with the predicted dose ($r=0.52$, $P=0.0004$). In the validation cohort, this algorithm accurately predicted warfarin dose than clinical data (0.92 vs. 0.51) in both high dose (0.91 vs. 0.77) and low dose (0.95 vs. 0.24) groups. This algorithm has increased sensitivity (0.98 vs. 0.52) and specificity (0.76 vs. 0.50) compared to clinical data. The rate of overestimation (0.24 vs. 0.50) and underestimation (0.02 vs. 0.48) was found to be significantly reduced with the use of algorithm.

Conclusions: The pharmacogenomic algorithm developed in the current study has high accuracy, high sensitivity and specificity in predicting the optimal dose of warfarin compared to clinical data.

Key Words:

Pharmacogenomics
Cardiovascular disease
Anticoagulation
Drug administration
Gene mutations